Weishu Zhang

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AI Infrastructure | Inference Systems, Workload Orchestration & Performance Modeling

Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences — *Expected May 2027*

Technical Skills

Languages: Python, C/C++, TypeScript/JavaScript, SQL
AI / ML Systems: PyTorch, torch.compile, vLLM, QLoRA, LLM APIs
Infrastructure: Kubernetes, Docker, gVisor, OCI, OpenTelemetry, CI/CD
Backend / Data: FastAPI, Node.js, React, PostgreSQL, Redis, REST APIs

Experience

Amazon
Software Development Engineer Intern — *Sep 2026 – Present*
- Developing distributed training and inference infrastructure for Amazon AI Studio.

ASML — San Jose, CA
Software Engineering, Test Infrastructure — *May 2026 – Aug 2026*
- Built RRI, an SQA regression-review system integrating Confluence, Jira, qTest, email, and compute-job status; automated review-page generation, metadata synchronization, and follow-up tracking.
- Designed an agent-driven test-plan lifecycle spanning case-to-plan generation, QA approval, asynchronous compute-job execution, and result/inspection writeback, with explicit review and execution stages.
- Built a human-reviewed knowledge-update loop that stages reusable session learnings for QA audit and promotes approved changes into the agent harness.

Ludus — Berkeley, CA
Co-Founder & Software Engineer, Cloud Agent Infrastructure — *Apr 2026 – Aug 2026*
- Implemented a Kubernetes-native control plane for stateful agent sandboxes, covering runtime templates, warm-pool provisioning, suspend/resume, and TTL-based lifecycle reconciliation.
- Isolated untrusted-code execution with gVisor RuntimeClasses, namespace-scoped RBAC, default-deny NetworkPolicies, and CPU, memory, and ephemeral-storage quotas.
- Packaged reproducible agent environments as immutable OCI images with PVC-backed workspaces and OpenTelemetry traces for sandbox lifecycle observability.

Berkeley Artificial Intelligence Research (BAIR) — Berkeley, CA
Undergraduate Research Assistant — *Mar 2026 – May 2026*
- Coordinated domain-expert task acquisition for the Agent Human Last Exam benchmark, translating expert proposals into cross-domain agent evaluation tasks.
- Built an LLM-assisted proposal workflow with structured generation, iterative refinement, and constraint checks to turn rough ideas into reviewable benchmark submissions.

University of California, San Francisco — San Francisco, CA
Software Research Assistant — *Aug 2025 – Jan 2026*
- Developed a long-context clinical information-extraction pipeline, decomposing records into schema-specific subtasks and combining output validation with self-consistency checks.
- Constructed domain-specific training data and fine-tuned Qwen with QLoRA; optimized vLLM inference through quantization and batched execution to address memory and throughput constraints.

Open Source

PyTorch | *torch.compile, Dynamo* — Commit
- Landed a CELU validation fix aligning torch.compile with eager-mode behavior for alpha=0; added regression tests for numerical parity, in-place mutation, and invalid inputs.

Selected Projects

mini-vLLM | *LLM Serving & Performance Modeling* — Code
- Built a CPU-only serving simulator with paged KV allocation, reference-counted prefix caching, copy-on-write, continuous batching, and recompute/swap preemption; added deterministic workloads and an interactive execution trace.
- Benchmarked KV capacity, throughput, and time-to-first-token (TTFT) under deterministic workloads; paged allocation fit 112 versus 16 sequences under a fixed 32,768-slot budget, with simulated compute.

GradeView / GradeSync | *Berkeley Course Infrastructure* — Live
- Refactored Berkeley course analytics into containerized React, Node.js, and Python services; centralized course data in PostgreSQL and added Redis caching/prewarming for repeated queries.
- Normalized score data across Gradescope, PrairieLearn, and iClicker with role-aware access for students and course staff.